Abdellah Arrouche

+213555293278 | Algiers, Algeria | abdellah.arrouche.03@gmail.com | Linkedin | Portfolio | Github

Profile

A graduate from The National Higher School of Computer Science (**French:** École nationale Supérieure d'Informatique, **ESI**), fascinated by Data Science, Information Visualisation and Business Intelligence. Curious, independent and rigorous, I am looking for employment opportunities.

Education

National Higher School of Computer Science (ESI) — Algiers, Algeria
Master's & State Engineer's Degrees in Intelligent Systems & Data Science — *Sep. 2021 – Jun. 2026*

Experience

Video Game Developer — Sep. 2025 – Jun. 2026
Internship at Laboratoire de Méthodes de Conception des Systèmes (LMCS) — *Algiers, Algeria*

- Refactal:Reimagined, an educational serious game for learning advanced Object-Oriented Programming, as a 2D puzzle-platformer built in Godot, integrating executable code directly into gameplay.
- Developed a custom code interpreter with lexical analysis, parsing, validation, and execution, allowing player-written code to trigger visible interactions with in-game objects.
- Implemented analytics and player-tracking infrastructure using asynchronous HTTP requests and a Supabase PostgreSQL backend, enabling gameplay, progression, and learning data collection.
- Evaluated the redesigned game with 52 students using gameplay analytics, pre/post knowledge quizzes, and the MEEGA+ model, achieving 4.22/5 satisfaction and 4.50/5 reported fun.

Stack: Godot 4.6, GDScript, PostgreSQL, Supabase

Data Analyst / Dashboard Developer — Sep. 2024 – Oct. 2024
Internship at Laboratoire de Méthodes de Conception des Systèmes (LMCS) — *Algiers, Algeria*

- Developed an administrative analytics dashboard for Refactal, a serious game designed to teach advanced Object-Oriented Programming concepts.
- Built interactive dashboards for monitoring player progression, level performance, engagement, room activity, quizzes, and key gameplay metrics.
- Integrated gameplay data from PostgreSQL into Metabase, providing visual analytics to support evaluation of player behavior and game design.
- Implemented concurrent access synchronization for multiple moderators managing the dashboard and game data.

Stack: PostgreSQL, Metabase, Docker, SQL

Technical Skills

Languages: Java, Python, C#, JavaScript, TypeScript, PHP, C, Pascal, SQL, R.
Web: React.js, TailwindCSS, Bootstrap, Django, Flask, FastAPI.
Databases: PostgreSQL, MySQL, SQLite, MongoDB.
Business Intelligance Software: Microsoft Power BI, Tableau, Metabase, Talend, Pentaho Data Integration.
Tools: Git, GitHub, Figma, Jira, Slack, ClickUp, VS Code, UML.

Soft Skills

- Curious, Independent, and Rigorous.
- Problem-solving, and attention to detail.

Projects

- **Refactal: Reimagined:** A 2D educational puzzle-platformer for learning advanced Object-Oriented Programming, built with Godot and featuring a custom code interpreter that executes player-written code in the game world.
- **NAS for PFSP:** A Neural Architecture Search (NAS) approach for solving the Permutation Flow Shop Scheduling Problem (PFSP), using a Variable Negihborhood Search and Genetic Algorithm Hybrid method.
- **Refactal Dashboard:** An administrative analytics dashboard for the Refactal serious game, providing insights into player progression, engagement, level performance, and gameplay activity through PostgreSQL and Metabase.
- **Scientific Articles Search Engine:** A web application for searching scientific articles using keywords and full-text search, built with ReactJS, Django, PostgreSQL, and ElasticSearch.
- **Bank Offers Comparison Website:** A responsive web platform for comparing banking services and fees in Algeria, featuring filtering, sorting, bank information, and side-by-side comparison using HTML, CSS, JavaScript, PHP and MySQL.

Languages

Arabic: Native | English: Fluent | French: Fluent